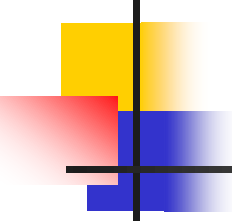


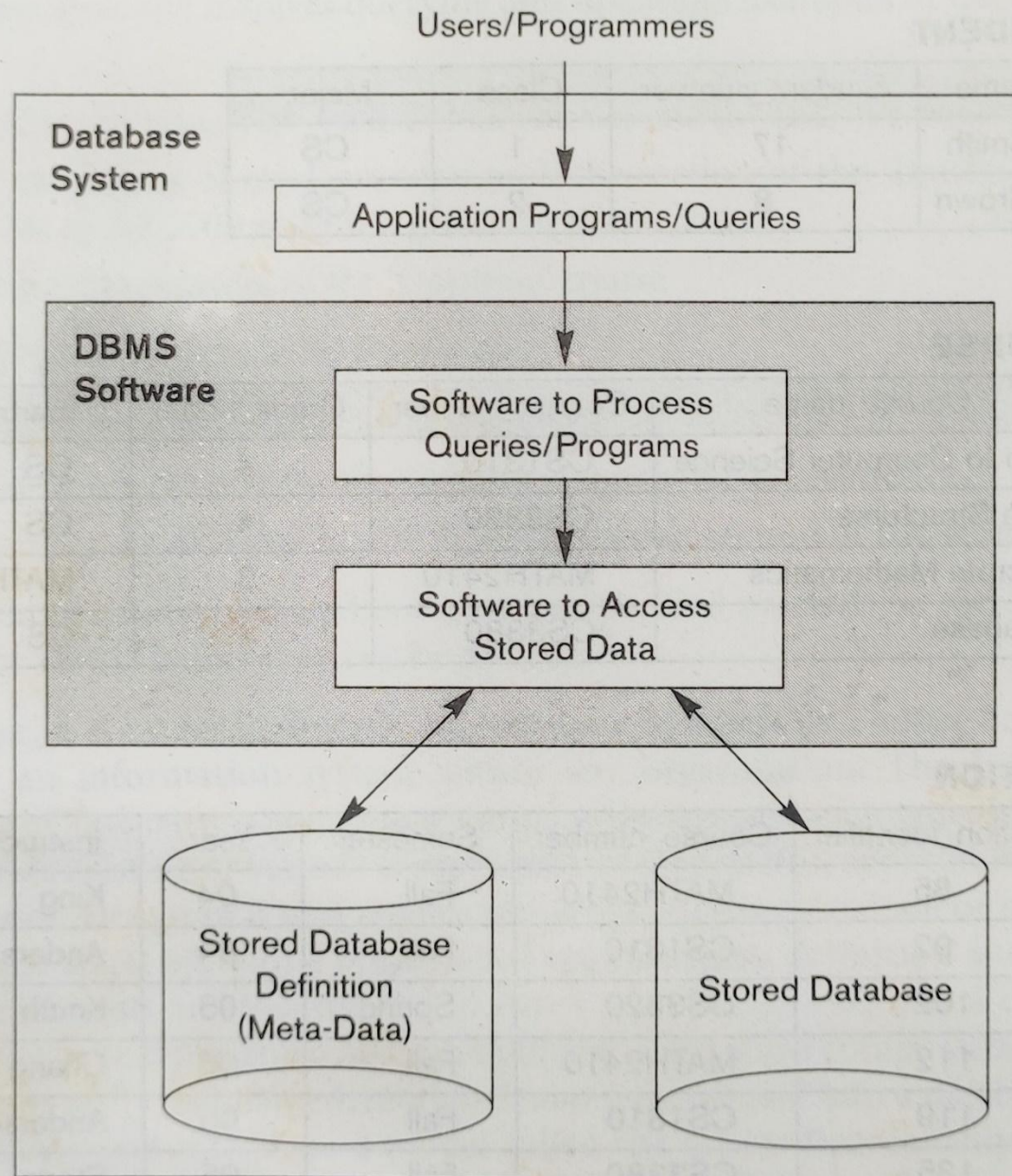
Introduction to Database Systems





What is Database?

- Essentially, a database is a set of data, or a collection of information. These data are related.



A Simplified
Database
System
Environment

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

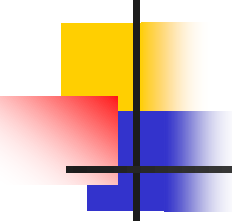
GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

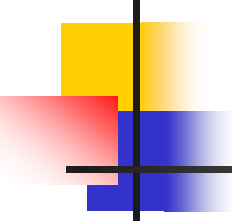
A database that stores student and course information



What's DBMS?

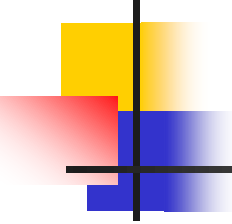
- DataBase Management System (DBMS)

is a collection of programs that enables users to create and maintain a database. The DBMS is hence a *general-purpose software system* that facilitates the processes of *defining*, *constructing*, and *manipulating* databases for various applications



Typical DBMS Functionality

- Define a database : in terms of data types, structures and constraints
- Construct or Load the Database on a secondary storage medium
- Manipulating the database : querying, generating reports, insertions, deletions and modifications to its content



Main Characteristics of the Database approach

Self-describing nature of a database system: A DBMS **catalog** stores the *description* of the database. The description is called **meta-data**). This allows the DBMS software to work with different databases.

Insulation between programs and data: Called **program-data independence**. Allows changing data storage structures and operations without having to change the DBMS access programs.

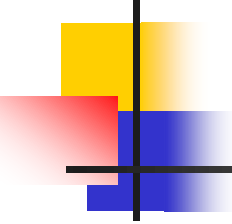
RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

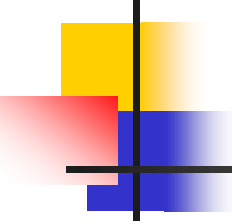
An example of a
Database catalog



Main Characteristics of the Database approach

Data Abstraction: A **data model** is used to hide storage details and present the users with a *conceptual view* of the database.

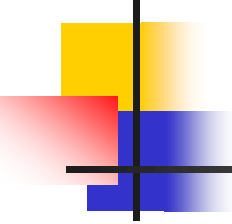
Support of multiple views of the data: Each user may see a different view of the database, which describes *only* the data of interest to that user.



Main Characteristics of the Database approach

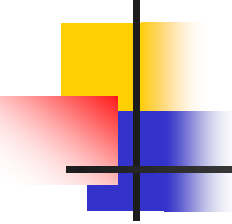
Sharing of data and multiuser transaction

processing : allowing a set of concurrent users to retrieve and to update the database. Concurrency control within the DBMS guarantees that each **transaction** is correctly executed or completely aborted. OLTP (Online Transaction Processing) is a major part of database applications.



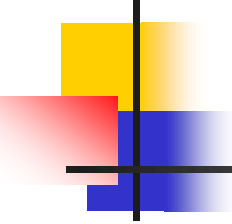
Database Users

Users may be divided into those who actually use and control the content (called "Actors on the Scene") and those who enable the database to be developed and the DBMS software to be designed and implemented (called "Workers Behind the Scene").



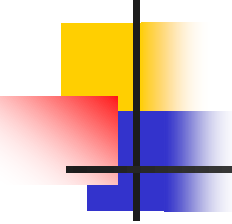
Database Users

- Actors on the scene
 - Data administrators : responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software, and hardware resources and controlling its use and monitoring efficiency of operation



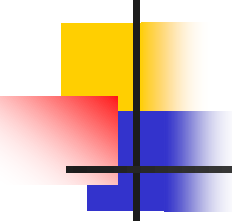
Database Users

- Database Designers : responsible for define the content, the structure, the constraints, and functions or transactions against the database. They must communicate with the end-user and understand their needs.
- End Users : they use the data for queries, reports and some of them actually update the database content.



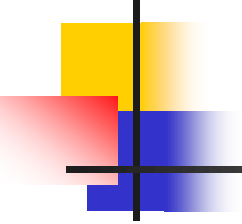
Categories of End-users

- **Casual** : access database occasionally when needed
- **Naïve or Parametric** : they make up a large section of the end-user population. They use previously well-defined functions in the form of “canned transactions” against the database. Examples are bank-tellers or reservation clerks who do this activity for an entire shift of operations.



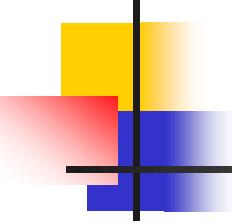
Categories of End-users

- **Sophisticated** : these include business analysts, scientists, engineers, others thoroughly familiar with the system capabilities. Many use tools in the form of software packages that work closely with the stored database.
- **Stand-alone** : mostly maintain personal databases using ready-to-use packaged applications. An example is a tax program user that creates his or her own internal database.



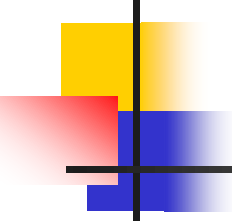
Advantages of Using the database Approach

- Controlling redundancy
- Restricting unauthorized access to data.
- Providing persistent storage for program Objects



Advantages of Using the database Approach

- Providing backup and recovery services.
- Providing multiple interfaces to different classes of users.
- Representing complex relationships among data.
- Enforcing integrity constraints on the database.



Reference Books:

- **Fundamentals of Database systems**, Fifth Edition.

Author: Elmasri and Navathe

- **Database system Concepts**,

Author: Abraham Silberschatz (Bell Laboratories), Henry F. Korth(Bell Laboratories) and S. Sudarshan (Indian Institute of Technology, Bombay).

Publishers: The McGraw-Hill Companies, Inc.